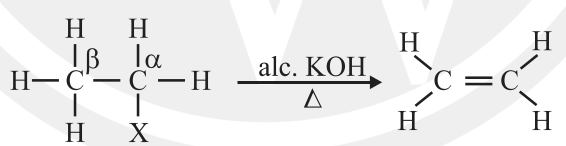
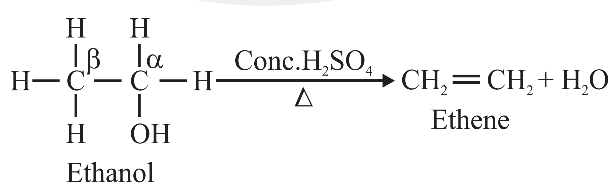
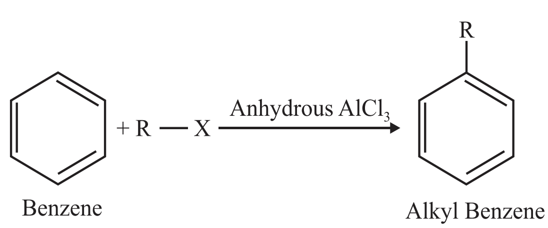
CHAPTER 10

Hydrocarbons

Sr. No.	Concept	Description	Other information
1.	**Important reactions:**		
	(a) Hydrogenation	$\underset{\text{Ethene}}{CH_2 = CH_2} + H_2 \xrightarrow{Pt/Pd/Ni} \underset{\text{Ethane}}{CH_3 - CH_3}$	Reagent = Pt, Pd, Ni
	(b) Wurtz reaction	$2RX + 2Na \longrightarrow R-R + 2NaX$	Reagent = Na, dry ether, X = Cl, Br
	(c) Decarboxylation	$R-COONa \xrightarrow[\text{Heat}]{\text{NaOH \& CaO}} R-H + Na_2CO_3$	Reagent = NaOH & CaO
	(d) Kolbe's electrolysis	$2RCOO - Na + 2H_2O \longrightarrow R - R + 2CO_2 + H_2 + 2NaOH$	
	(e) Dehydrohalogenation		X = Cl, Br, I Reagent = alc. KOH
	(f) Acidic dehydration of alcohols.		Reagent = Conc. H_2SO_4
	(g) Friedel Craft's reaction (i) Friedel Craft Alkylation		Reagent = Anhydrous $AlCl_3$ X = Cl, Br, I

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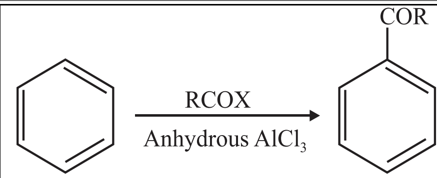
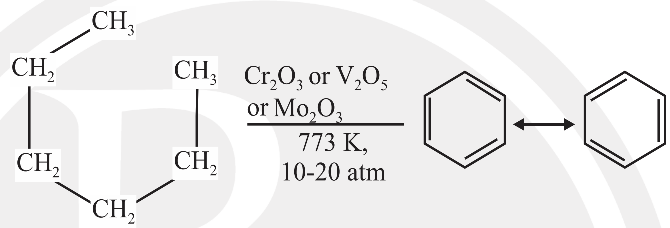
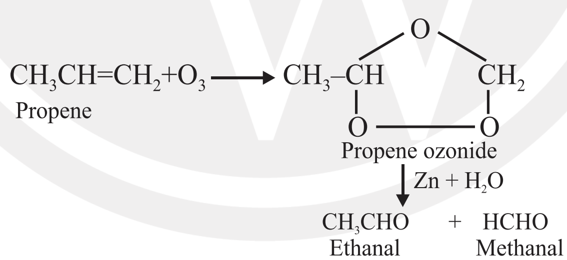
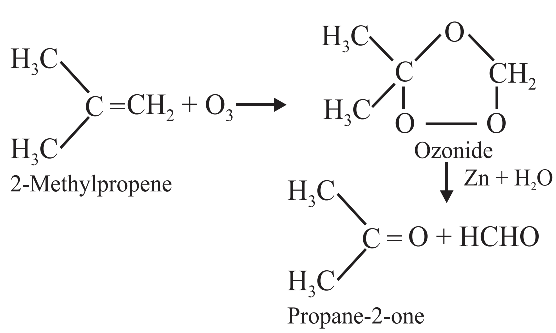
	(ii) Friedel Craft Acylation	 Benzene	Reagent = Anhydrous $AlCl_3$ X = Cl, Br, I
	(h) Pyrolysis:	$C_6H_{14} \xrightarrow{773\ K} C_6H_{12} + H_2$ $\phantom{C_6H_{14}} \xrightarrow{773\ K} C_4H_8 + C_2H_6$ $\phantom{C_6H_{14}} \xrightarrow{773\ K} C_3H_6 + C_2H_4 + CH_4$	High temperature is required.
	(i) Aromatization		Reagent = Cr_2O_3 or V_2O_5 or Mo_2O_3
	(j) Isomerisation	$CH_3(CH_2)_4CH_3 \xrightarrow{\text{Anhy. } AlCl_3/HCl}$ *n*-Hexane $CH_3\underset{\displaystyle CH_3}{\underset{\vert}{C}H}\text{–}(CH_2)_2\text{–}CH_3 + CH_3CH_2\text{–}\underset{\displaystyle CH_3}{\underset{\vert}{C}H}\text{–}CH_2\text{–}CH_3$ 2-Methylpentane, 3-Methylpentane	Reagent = Anhy. $AlCl_3$/HCl
	(k) Ozonolysis	 	Aldehyde and ketone both can be prepared by ozonolysis

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2.	**Physical properties of alkane**		
	(a) Physical state	(i) C_1 to $C_4 \rightarrow$ gases. (ii) C_5 to $C_{17} \rightarrow$ liquids. (iii) Higher than $C_{18} \rightarrow$ solids. (iv) Colourless, odourless, insoluble in water, dissolve in non-polar solvents.	
	(b) Boiling point	(i) For the same alkane $\rightarrow$ boiling point decrease with branching	B.P. $\propto$ Mwt.
	(c) Reactivity of alkane	Reactivity of alkane towards free radical halogenation is $\propto$ stability of free radical $C_6H_5-CH_3 > CH_2=CH-CH_3 > (CH_3)_3CH$ $> CH_3-CH_2-CH_3 > CH_3-CH_3 > CH_4$	
	(d) Reactivity of halogens	Reactivity of halogen towards free radical substitution $F_2 > Cl_2 > Br_2 > I_2$	
	(e) Knocking tendency	Knocking tendency of petroleum as fuel decrease with increase in side chain. Straight chain > Branched chain Knocking tendency is in the order. Olefin > cycloalkane > aromatic	
3.	**Physical properties of alkene**		
	(a) Physical state	(i) C_1 to $C_3 \rightarrow$ gases. (ii) C_4 to $C_{14} \rightarrow$ liquids. (iii) Higher than $C_{14} \rightarrow$ solids. (iv) Colourless, odourless, insoluble in water, fairly soluble in non-polar solvents.	
	(b) Boiling point	(i) Straight chain alkenes $\rightarrow$ higher boiling points than isomeric branched chain compounds.	B.P. $\propto$ Mwt.
	(c) Order of reactivity of olefins	Order of reactivity of olefins for hydrogenation $CH_2=CH_2 > R-CH=CH_2$ (Reverse of stability)	

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4.	**Physical properties of alkyne**		
	(a) Physical state	(i) C_1 to $C_3 \rightarrow$ gases. (ii) C_4 to $C_8 \rightarrow$ liquids. (iii) Higher than $C_8 \rightarrow$ solids. (iv) Colourless, odourless, insoluble in water, soluble in non-polar solvents.	
	(b) Boiling point		B.P. $\propto$ Mwt.
	(c) Melting point	Melting point depends upon symmetry.	
5.	**Physical properties of aromatic hydrocarbons:**		
	(a) Physical state	(i) Non-polar molecules (ii) Colourless liquids or solids, immiscible with water, miscible with organic solvents, burn with sooty flame,	
	(b) Boiling point		B.P. $\propto$ Mwt.
	(c) Melting point		Melting point $\propto$ symmetry